

USER GUIDE FR3D MK3 — Chapters 1, 2, 3, 5

Source: `USER GUIDE\_FR3D\_MK3\_EN.pdf`

Clean extract for Google Docs / GitHub (standalone MK3S+ understanding).

Chapters included: **1, 2, 3, 5** (main screen), **6** (operation recommendations). Chapter **4** omitted.

1. Project objective

Build a filament extrusion and spooling platform with digital control, telemetry and traceability, based on the MK3S+ with FR3D Addon firmware operating in standalone mode from the machine LCD.

Documentation scope. This document is directed exclusively at the **MK3S+ with FR3D Addon** operating in **standalone** mode (LCD and onboard firmware control). It does not cover Raspberry Pi gateway pairing or the browser-based operations UI.

Diameter sensor hardware. The MK3S+ can use either a **Hall-effect diameter sensor** (INFIDEL-style printed assembly, SS495A on pin **A3**) or an **optical diameter sensor** based on **ARTME 3D**. Calibration on the LCD is the same for both. Printed parts for the Hall-effect head:

`documnetation/diameter\_sensor\_infidel/Hall\_Sensor\_Parts.stl` (multi-body STL — split into individual parts in the slicer before printing).

- Real-time control of the extruder and spooler.
- Capture and analysis of process data in real time and in batch mode (CSV formats).
- Interface-driven operation with multilingual support (ES/EN).
- Architecture ready for further automation and quality improvements.

2. Addon FR3D

- Modified MK3S+ firmware for USB serial communication with the Python core on a Windows

PC.

- Python core synchronized with USB serial ports and control/telemetry logic for integration

with Addon FR3D - MK3S+.

- Structured configuration, event logging, analysis and management in CSV.
- Coordinated operation flow with MK3 firmware .
- The two main functions are the Caliper and the diameter predictor

2.1 Diameter Meter Caliper

The Caliper is only for calibrating the diameter gauge and ensuring it measures correctly. Calibration is only done once during assembly. Then, the offset is used to adjust the zero point if it's misaligned (as is the case with all calibrators on the market that have a zero adjustment). Specifically, this function is only for calibrating the gauge so that the display shows the actual measurement, where the lower limit is 1.5 mm and the upper

limit is 2.5 mm. The gauge will always indicate the actual measured value between 1.5 and 2.5 mm. Outside these limits, it displays 1.5 or 2.5 mm due to a mechanical issue related to the physical sensor.

2.2 Diameter Predictor (FR3D MK3S+)

2.2.1) What is the Predictor? The Diameter Predictor is a closed-loop assistant that runs on the MK3S+ FR3D firmware. It uses the **diameter sensor** calibrated in the previous step to keep the extruded diameter close to the target (typically 1.75 mm, within the parameters given to the predictor). The predictor stays inactive until calibration is valid (``CALV=1``). Every ~10 seconds, the predictor analyzes the data and, if correction is required and safety conditions are met, the algorithm adjusts:

1) Extruder speed (E): This is the fastest response to diameter errors. 2) Nozzle temperature (T): When RPM alone is insufficient, the predictor switches to a thermal correction strategy.

2.2.2) Correction (Safety Gates)
The predictor does not act unless, among other checks:

- The predictor is in automatic mode.
- The extruder is running ($ES = 1$).
- The current temperature is close to the setpoint ($|T_{act} - T_{tgt}| \leq$ the maximum configured

temperature).

- Spolling is set to automatic.
- The diameter signal is stable, with no electrical noise (medium debounce: large jumps

require confirmation using Med jump/Med confirm).

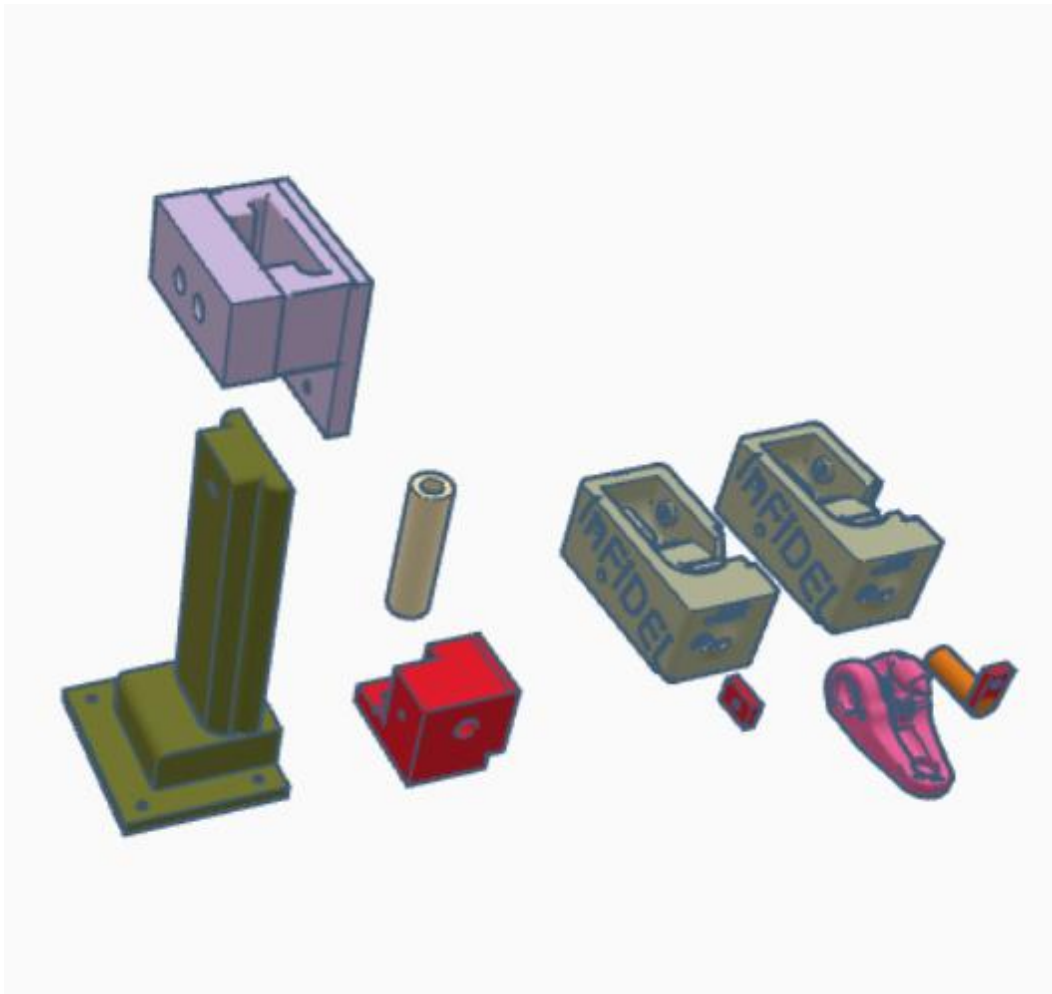
- The proposed R and T values are maintained within R_{min}/R_{max} and T_{min}/T_{max} . (This determines the minimum and maximum adjustment limits of MK3S+.)

- What is the target diameter and what margin of error does it tolerate before taking action?

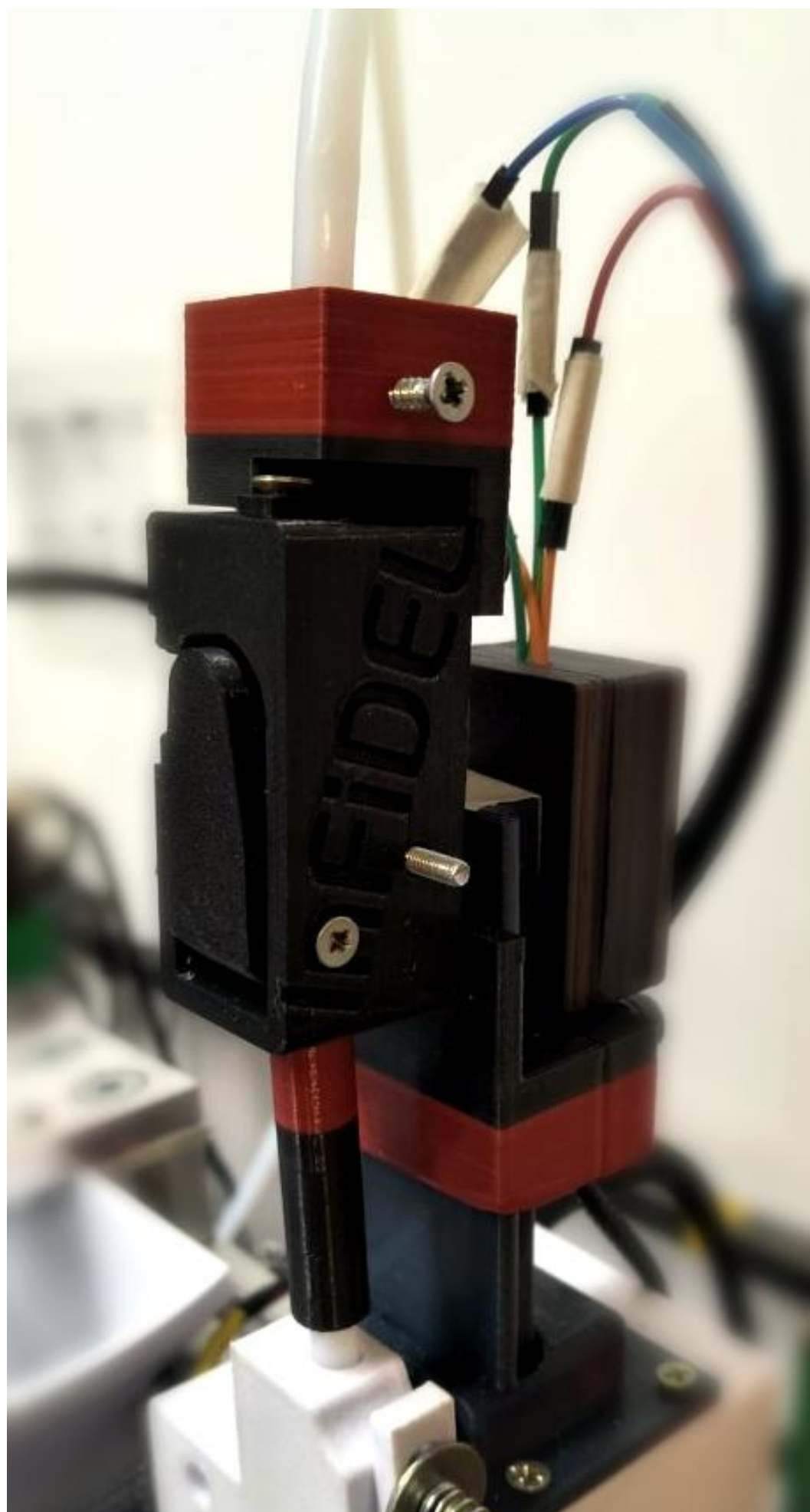
etc..

3. Diameter Hardware connection

Sensor options. Hall-effect (INFIDEL-style, SS495A on A3 with EMI filters below) or optical (ARTME 3D). For Hall-effect mechanics, print the nine bodies from ``Hall_Sensor_Parts.stl``, assemble with roller/magnet/SS495A, and mount on the filament path before the puller.



INFIDEL printed parts



INFIDEL assembled on MK3S+

Recommended Technical Note: Noise Reduction Filter for the SS495A (Hall-effect option). To protect the highly sensitive SS495A linear Hall-effect sensor from high-frequency electromagnetic interference (EMI) caused by stepper motors and heating elements, a two-stage filtering strategy is implemented. This will be done in two stages: one for decoupling the power supply and another RC low-pass filter stage for the analog signal.

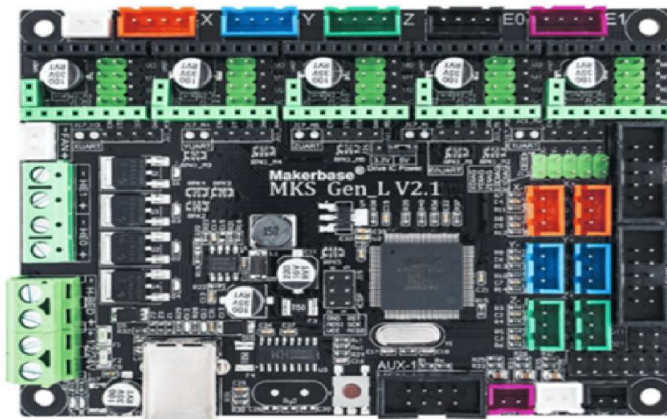
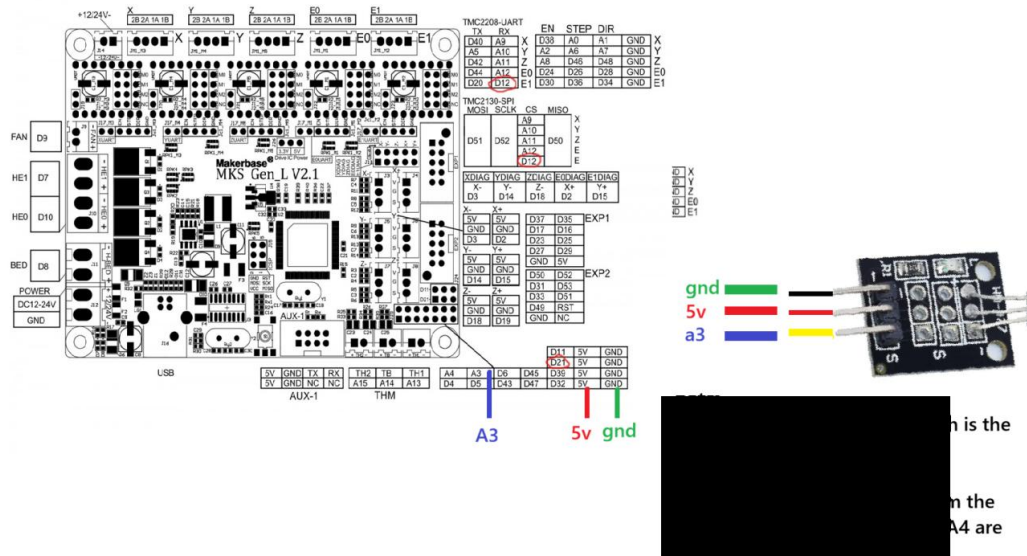


Figure p5

Stage 1 - Power Supply Decoupling (located 4 cm from the sensor). Components:

- One 0.1 μF ceramic capacitor (labeled 104)
- One 4.7 μF electrolytic capacitor.

Wiring Setup: Connect the positive (+) lead of the 4.7 μF electrolytic capacitor (the longer leg) to one leg of the 0.1 μF ceramic capacitor. Solder this connection directly to the 5V (Vin) line. Connect the negative (-) lead of the electrolytic capacitor (the shorter leg aligned with the strip on the body) to the remaining leg of the ceramic capacitor. Solder this connection directly to the GND line. Note: The sensor's output (data) wire completely

bypasses this block and continues along the line. Stage 2 - Signal Data Filtering (RC Low-Pass Filter):
Components:

- One 1kΩ resistor
- One 10nF ceramic capacitor (labeled 103).

Wiring Setup: Resistor Placement: Cut the signal (output) wire coming from the sensor. Solder the incoming wire to one end of the 1kΩ resistor. Connect the other end of the resistor directly to the microcontroller's analog input pin. Capacitor Placement (Critical): Place the 10nF ceramic capacitor as close as possible to the microcontroller pin (ideally less than 4 cm or directly on the board). Solder one leg of the capacitor to the analog input pin (the same junction where the resistor ends) and the other leg directly to the microcontroller's GND pin.

1. MK3S + FR3D main screen (1) Displays current filament diameter (2) Displays average filament diameter (from the last 10 seconds - with 20 samples) (3) Displays prediction mode: A = automatic adjustment, M = manual adjustment

5.1 Configuration

You can configure from the Python / web console or from the MK3 LCD: **Control → AddonFR3D**.

5.1.2 Diameter Sensor (LCD)

Path: **Menu → Control → AddonFR3D → Diameter Sensor**

Menu item	Role
Pattern Diameter	Shows Now (active mm preset) and Set for the *other* preset only
Need Calibrate!	Only when calibration is incomplete (`CALV=0`). Opens the Capture list
CALIBRATE	Only when calibrated (`CALV=1`). Contains Capture Lo / Mid / Hi
Offset	Fine diameter compensation (DOFF), typically -0.20...0.20 mm

Presets (Pattern Diameter)

Preset	Lo / Mid / Hi (mm)
Wide (A)	1.5 / 1.7 / 2.0
Standard (B, factory default)	1.7 / 1.75 / 2.0

Changing pattern with **Set**:

2. Invalidates calibration (`CALV=0`) and clears the three Capture points.
3. Sets **OFFSET (DOFF) to 0** (does not carry over bias from the previous preset).
4. Shows **New Setting OK / Need Calibrate!!! / Click To calibrate** — click opens Capture for the first point.

5.1.2.1 Diameter meter calibration (Capture)

Use physical calibration patterns that match the **active** preset labels on the LCD.

5. Open **Need Calibrate!** or **CALIBRATE**.
6. Place the matching physical pattern under the diameter sensor.

7. Select the Capture row (e.g. Capture 1.75 or Capture 1.50 depending on preset).
8. Wait at least 5 seconds for a stable ADC reading.
9. Click the encoder to save. Repeat for all three points until `CALV=1`.

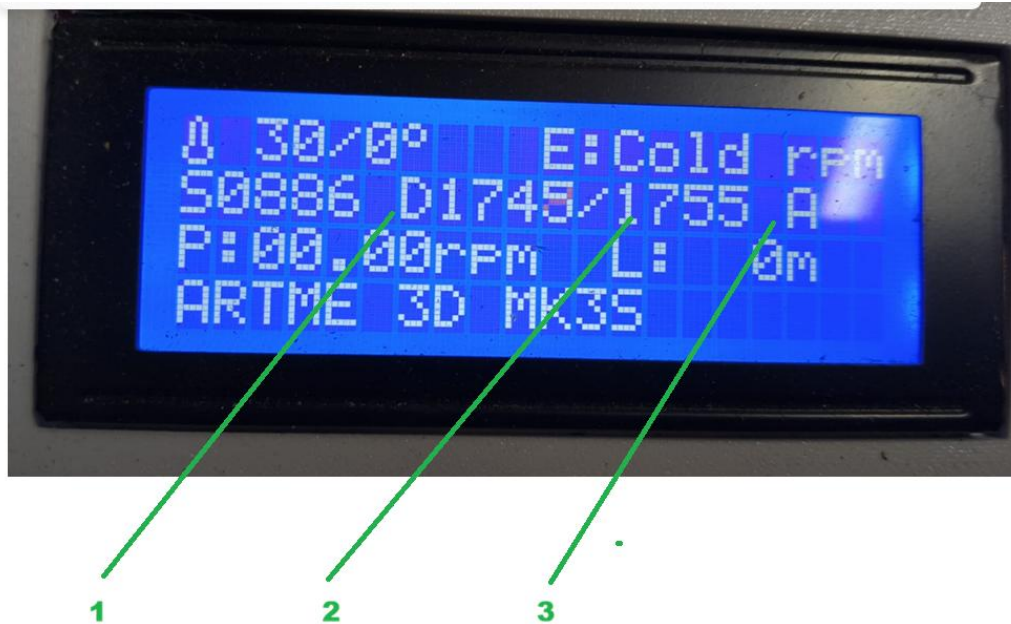


Figure p8

5.1.2.2 Calibration verification

10. Place a known mid pattern (e.g. 1.75 mm when on the standard preset) under the diameter sensor.
11. Wait at least 15 seconds and return to the main screen.
12. On the second LCD row you see `DXXXX/YYYY` (instant / average $\times 1000$). For 1.75 mm expect about **1750**.

5.1.2.3 OFFSET compensation

If the screen shows e.g. 1720 instead of 1750:

13. Go to **AddonFR3D → Diameter Sensor → Offset**.
14. Apply the difference (example: $1750 - 1720 \rightarrow 0.030$ mm) and click to store.
15. Confirm the main screen reads near `D1750/1750`.

Note: After a **Pattern Diameter → Set** change, Offset is reset to **0**; recalibrate the three Captures before tuning Offset again.

LCD screenshots — Diameter Sensor menus:

Diameter Sensor (Pattern Diameter / CALIBRATE / Offset) and **Control → Diameter Sensor**:

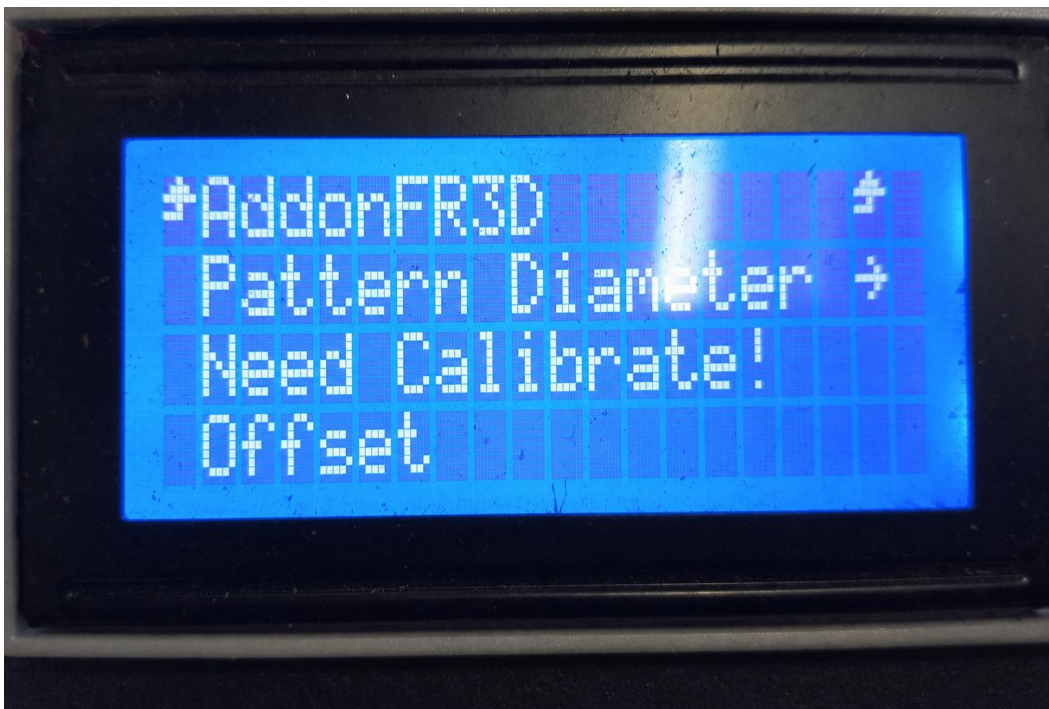


Figure p10



Figure p10

When calibration is incomplete ('CALV=0'), the middle item is **Need Calibrate!** instead of **CALIBRATE** (simulated LCD — same layout as a live panel):



Need Calibrate

Pattern Diameter (Now / Set) and **CALIBRATE** Capture list (labels follow the active preset):



Figure p11

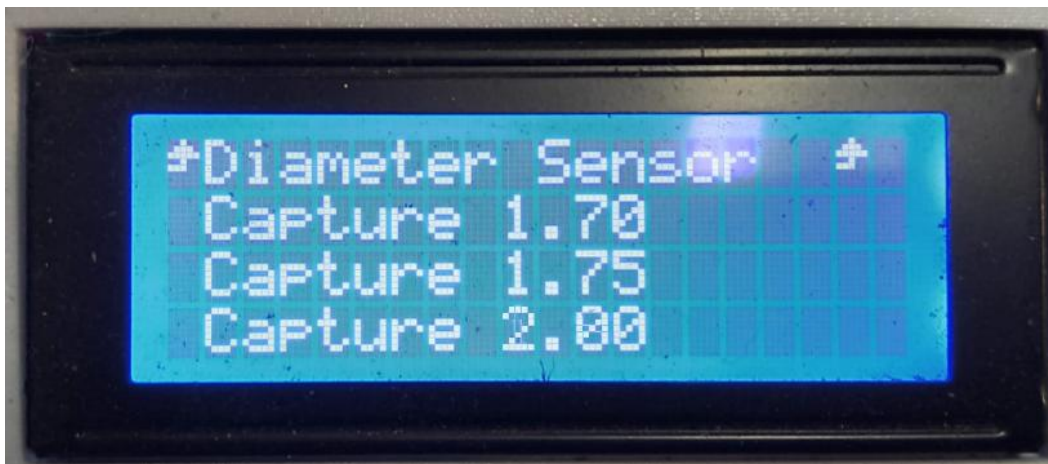


Figure p11

5.2 MK3 Predictor Diam

Predictor Auto ON/OFF is on the **Main menu**, at the same level as **Automatic / Manually Pulling** (not inside Predictor Parm's).



Figure p12 — Main menu: Predictor Auto

Predictor Parm's (Settings → Predictor Parm's) starts with limits — there is **no Mode line** here anymore:



Figure p12 — Predictor Params: Emin / Emax / Tmin

Menu Predictor Diam (2/7) Menu Predictor Diam (3/7)



Figure p13

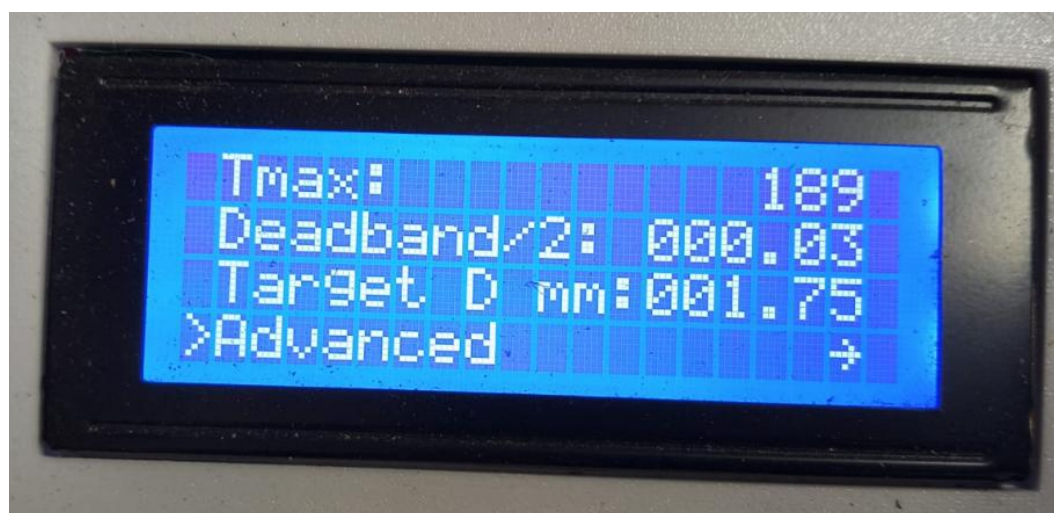


Figure p13

Menu Predictor Diam (4/7) Menu Predictor Diam (5/7)

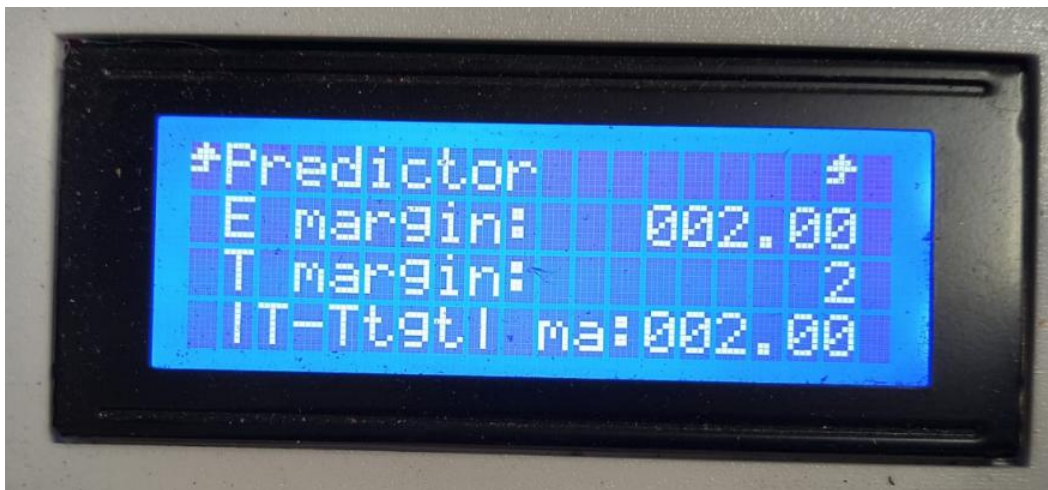


Figure p14



Figure p14

Menu Predictor Diam (6/7) Menu Predictor Diam (7/7) **Predictor Auto** — On the Main menu (same level as Automatic / Manually Pulling): **ON** applies real-time corrections within configured limits; **OFF** computes/suggests only (status letter **A** / **M** on the home screen). Emin — Lower RPM limit allowed for extrusion-speed corrections. Prevents the predictor from reducing speed too much and compromising flow continuity. Emax — Upper RPM limit allowed for speed corrections. Prevents over-acceleration and saturation-driven oscillations in the extruder. Tmin — Lower thermal limit of the assistant working range (it does not change the base RTF setpoint by itself). Used to bound temperature correction decisions.



Figure p15



Figure p15

Tmax — Upper thermal limit of the assistant range. Keeps corrections inside a safe zone for material/process.

Deadband/2 mm — Half of the deadband around target diameter. If diameter stays within target $\pm$ band, predictor does not correct. Larger bands = fewer corrections; smaller bands = more sensitivity.

Target D mm: — Target filament diameter the system tries to maintain. All adjustment and deadband logic is computed against this value.

=====Advanced=====

Advanced-Wait after T change — Number of cycles/fusions the predictor waits after applying a temperature change before correcting T again. Reduces thermal-inertia rebound.

Advanced-E margin to switch to T (rpm) — Hysteresis that authorizes transition from RPM correction to temperature correction. Avoids overly frequent strategy switching.

Advanced-T margin to switch to R ($^{\circ}$ C) — Hysteresis to return from T correction to R correction. Improves switching stability between channels.

Advanced-|Tact-Ttgt| max $^{\circ}$ C — Maximum allowed difference between actual and target temperature to enable predictive correction. If exceeded, the predictor is inhibited to avoid correcting in unstable conditions.

Advanced-T settle N — Predictor window (number of samples per cycle). Higher N filters noise and stabilizes; lower N speeds up corrections but may overfit. Suggested starting point: N=3.

Advanced-CYCLE — CYCLE — Telemetry and predictor cycle period, in seconds. Allowed values: 5 or 10 (default 10). Defines how often the MK3 closes a sampling window and performs a diameter predictor evaluation. With 10 s, the system filters noise better and corrections are more stable. With 5 s, it reacts faster to diameter changes, at the cost of

greater sensitivity to sudden variations. Advanced-Med jump mm (DIAMJDB) — How much the ~10 s rolling median diameter must change versus the last published median to treat it as a “large jump” that is NOT applied immediately; the MK3 keeps the previous published median while awaiting confirmation in the

following window before updating predictor logic and CSV. Higher values tolerate transients better (e.g. inserting/cutting filament); lower values react faster but trigger debounce more often. It does not fix systematic sensor calibration — use DOFF / diameter-sensor Offset for bias. Advanced-Med confirm mm (DIAMJPM) — After a pending large jump crosses “Med jump”, firmware requires that the raw median of the next window matches the pending candidate within $\pm$ this tolerance (absolute) before publishing; otherwise the hold can continue/reject. It is the confirmation slack between successive windows: higher accepts more easily; lower demands a cleaner match. Advanced-DIAMDEBUG — MK3 CSV format only: appends lightweight diagnostic columns to see FIFO floor rejects (corr_drop_n), median debounce holding (corr_jump_hold, jump_pending_mm), and raw vs accepted FIFO fill (fifo_n_raw / fifo_n_used). Handy during tuning; commonly OFF during normal runs for shorter rows. Does not alter predictor correction logic. The CSV is sent to Python control to process. Advanced-dE min — Minimum RPM correction step per cycle when acting on R. Prevents corrections that are too small to have practical effect. Advanced-dE max — Maximum RPM correction step per cycle. Limits abrupt changes to avoid mechanical/flow instability. Advanced-kSpan E — E gain associated with distance to the diameter band (span component). Higher values respond more strongly when outside band. Advanced-kErr E — E gain associated with absolute diameter error (error component). Increasing it speeds R correction, but can induce overshoot. Advanced-dT min C — Desired/base thermal change per cycle when channel T decides to act. Represents the minimum thermal correction intent. Advanced-dT max C — Maximum allowed temperature change per correction/cycle. In practice: applied dT = min(desired dT, max dT). Keep low (e.g., 1 °C) due to thermal inertia. Advanced-kSpan T — Thermal gain by distance to band (span component of T channel). Advanced-kErr T — Thermal gain by diameter error (error component of T channel). High values correct faster but may cause thermal oscillation.

5.2.1 Recommended Settings

Predictor ON/OFF Diamdebug: ON

Modo Manual/automatic Target Diam: 1.75 Deadband/2: 0.03 Med Salto m: 0.15 Med conf mm:0.06 Tem match: 2 E min:9 E max: 32 T min :174 Tmax:189 CYCLE: 10 dE min: 0.05 dE max:0.6 kSpan E: 10 kErr E:15 dT min:0.2 dT max:1 kSpan T:15 kErr T:55 E margin: 2 T margin: 2 T settle N: 3 Window N: 3

6. Operation recommendations

Objectives

Steps and recommendations for extrusion with **automatic predictive diameter control** (MK3S+ FR3D Addon, standalone LCD).

Step 1 — Prepare (Predictor Auto OFF)

- Set **Predictor Auto: OFF**.
- Review diameter-sensor calibration (Pattern Diameter / CALIBRATE on LCD; Hall-effect INFIDEL or ARTME 3D optical).
- Place shredded material in the hopper.
- Make sure the material is dry.

Step 2 — Base setpoints

Go to **Settings** and assign:

- Temperature
- Extruder RPM
- Fan

Step 3 — Start extrusion

Once the **target temperature** is reached, start extrusion.

Step 4 — Thread and enable pulling

Thread the extruded filament and set **Pulling (spooler) to Automatic**.

Step 5 — Enable Predictor Auto

When automatic pulling is stable and the diameter on the LCD is close to the target (shown as `Dxxxx` / average), set **Predictor Auto: ON**.

From this point, the prediction algorithm applies automatic diameter corrections each operating cycle.

Step 6 — Predictor reaches temperature / RPM limits

If the predictor saturates against its configured **temperature** and **RPM** limits (`Tmin`/`Tmax`, `Emin`/`Emax`), stop relying on further predictive corrections and apply the operator tips below (from in-app Help: Tips & tricks).

Filament too thin

When average diameter (`Dprom`) is below the acceptable window, work through these steps in order:

16. **Nozzle FAN** — Manually increase nozzle fan speed (F) so the filament stretches less under tension before solidifying.
17. **Mechanical distance adjustment** — Manually decrease the physical sensor distance (S), moving it closer to the nozzle to give the molten plastic less stretch time.
18. **Physical check and calibration** — Inspect internal wear of the nozzle orifice. Verify the diameter sensor is calibrated and not reporting false readings.
19. **Batch recipe adjustment** — Decrease recycled material percentage (–T%) and increase virgin material percentage (+V%) to reduce melt fluidity.

Filament too thick

When average diameter (`Dprom`) is above the acceptable window, work through these steps in order:

20. **Nozzle FAN** — Manually reduce nozzle fan speed (F) so the filament can stretch under tension before solidifying.
21. **Mechanical distance adjustment** — Manually increase the physical sensor distance (S), moving it away from the nozzle to give the molten plastic more stretch time.
22. **Physical check and calibration** — Inspect internal wear of the nozzle orifice. Verify the diameter sensor is calibrated and not reporting false readings.

23. **Batch recipe adjustment** — Increase recycled material percentage (+T%) and decrease virgin material percentage (–V%) to increase melt fluidity.